

1 Introduction

Naming scheme for ALICE Configuration in Run 3, v.0.1 . The aim of this document is to define a naming scheme for all object in t CTP configuration from detector to trigger classes in ALICE. The configuration of Central Trigger Processor (CTP) is closely connected with the naming of trigger classes. So the document in parallel with defining the **Codes** for trigger entities defines the entities itself in hierarchical way. The document is based on Run1/Run2 proposal [1, 2, 3].

2 General scheme

The basic units:

- a detector: detector
- an input: trigger input
- a L0 function: a logical function of the trigger inputs (maximum 4 ?)
- a descriptor: a list of the trigger inputs (maximum 24 ?) and L0 functions (maximum 2 ?), it acts as logical AND of inputs and L0 functions in the list
- a detector cluster: a list of detectors, defining which detectors are to be read-out together (common busy, i.e. the readout rate is determined by the slowest detector in the cluster)
- a bunch crossing mask, defining for which bunch crossings in the orbit the trigger is activated
- a class: descriptor + cluster + bunch crossing mask

In the present scheme, the information on these elements is explicitly en**Co**ded in the class **Code**. The class **Code** is formed as follows:

C<descriptor **Code**>-<bunch crossing mask **Code**>-<detector cluster **Code**>

Together with class **Code** it would be useful to assign each class a name (alias) which summarise its physics content (e.g. MinimumBias). The list of descriptor, bunch crossing mask, detector cluster **Codes** ("List of ALICE trigger class **Codes**") is maintained by the trigger menu task force (?). It is updated by the addition of new **Codes** as needed. Elimination/reuse of old **Codes** is not foreseen at the moment. The list has a version number formed as follows: yy-mm-dd-vv, such as: 08-06-30-00. The last two digits vv are foreseen in case more than one version is created the same day.

3 Detectors

Detectors names in clusters are according to O2 naming scheme:

ITS TPC TRD TOF PHS CPV EMC HMP
MFT MCH MID ZDC FT0 FV0 FDD ACO CTP BPM
BPM - Beam Position Monitor, not in O2.

Exception is the naming of CTP inputs where one letter detector codes as in Run2 are used 4

4 Inputs

Code: n<DAB> same convention as in run2

- n indicates the trigger input level (M,0, or 1),
- D is one letter detector name as defined in run2,
- AB is more detailed description of input.

Definition: n<DAB> Det Level Signature InpNum

- V : FV0
- T : FT0
- U : FDD
- O : TOF
- E : EMC
- D : EMC
- P : PHS
- H : HMP

5 LM/L0 function

LM/L0 function is any logical function of N inputs. *N to be discussed from the physics point of view and firmware restrictions.* LM/L0 **Code** has maximum 16 chars and syntax:

Code: n<FUNName>

Definition: 0<FUNName> logical expression of 4 inputs

In logical expression following operators are used:

| logical OR

& logical AND

$\wedge$ exclusive OR

$\sim$ negation

() for grouping operations

6 Descriptor

Each descriptor is defined as the list of LM/L0/L1 input functions **Codes** and input **Codes**. It is logical AND of the components in the list. Descriptor **Codes** start with the letter D and have a maximum length of 16 characters.

Code: <DName>

Definition: <DName> n<FUNName1> n<FUNName2> n<DETName1> ... n<DETName2>

...

In case of descriptors defined by a single trigger input, the descriptor code consists of the letter D followed by the trigger input code.

7 Bunch Crossing Mask Code

The bunch crossing mask defines for which bunch crossings (out of the 3564 in the orbit) the specific trigger class is enabled. The complete definition of the mask consists of a sequence of 3564 bits. For the purpose of offline use, however, the bunch crossings can be subdivided into a limited series of categories for which the action will be similar. Each category is indicated by a letter. The bunch crossing mask **Code** consists of the string of all the letters corresponding to the categories of bunch crossings for which the trigger class is active, in alphabetical order. e.g.: there are 4 bunch crossing types:

- B - A bunch and C bunch intersect in ALICE - two filled bunch crossings,
- A - filled bunch from side A but empty bunch from side C,
- C - filled bunch from side C but empty bunch from side A,

- E - empty from both sides

However for complicated beams we can use 'mixed masks', therefore Bunch Crossing Mask **Code** is

Code: <BCMName>

Definition: <BCMName> mask_expression

Where Name describes the masks content, if possible via the bunch crossing types. Maximum 16 chars. mask_expression defines 3564 bits of the mask. L denotes valid BC, H denotes vetoed BC. The number of repetitions (1..3564) can be before L/H. () can be used to define the repetition of the bit sequence. Examples: 1H3563L -all the BCs valid but the first one 1782(HL) -alternating pattern i.e. all the odd BCs (1,3,...,3563) are vetoed

8 Detector Cluster Code

Indicates the cluster of detectors which are read out in correspondence of the trigger class.

Code: <CLTName> (maximum 16 chars)

Definition: <CLTName> list of detectors

e.g. CLTALL FV0 ITS ... TPC

9 Trigger Class

Code: C<descriptor **Code**>-<bunch crossing mask **Code**>-<detector cluster **Code**>

Definition: C<descriptor **Code**>-<bunch crossing mask type>-<detector cluster **Code**> DName
CLTName BCName1 BCName2 Downscaling

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DOWNSCALES [clads]

list of downscaling

MASKS [bcm]

list of masks

GENS

list of generators

DESCRIPTORS (can be missing)

list of descriptors

LTG (= detector, new in RUN3 - needed for e.g. HBaccept rate ...)

list of detectors

CLUSTER [cluster]

list of classes

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VERSION

RUN runnumber

INPUTS

list of inputs

MASKS (can be missing)

list of masks

GENS

list of generators

DESCRIPTORS (can be missing)

list of descriptors

LTG (= detector, new in RUN3 - needed for e.g. HBaccept rate ...)

list of detectors

CLUSTER

list of classes

References

- [1] ALICE Technical Design Report of the Trigger, Data Acquisition, High-Level Trigger and Control System, CERN-LHCC-2003-062, ALICE TDR 10
- [2] v. 0.1 F Antinori, June 2008
- [3] v.0.2 R. Lietava, 26 January 2009